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Hierarchical VLMs Planning and Control with Ambiguity resolution

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Outline

1. Planning Layer

Neuro-Symbolic Planning architecture

VLM Architectures

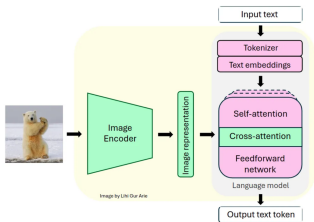


Figure: llama3 VLM architecture

Image + Text $\longrightarrow$ VLM $\longrightarrow$ Text Output

Language Encoder (Transformer)

- ▶ Encoders transform sequences into numerical representation called **embeddings**.
- ▶ A **self-attention mechanism** allows focus on important tokens.
- ▶ **Decoders** output the most statistically probable output.

Vision Encoder (CNN / ViT)

- ▶ Typically utilizes Convolutional Neural Networks (CNNs) or Visual Transformers (ViTs).
- ▶ A ViT processes an image into patches and treats them as sequences, akin to tokens.

VLM Training methodologies

It is the mechanism that allow fusing information from both vision and language encoders so to correlate images with text and make decision on the 2 modalities together:

- ▶ Contrastive learning: uses joint or shared embedding space. The VLM is trained on datasets of image-text pairs and learns to minimize the distance between embeddings of matching pairs and maximize it for nonmatching pairs.
- ▶ Masking: models learn to predict randomly obscured parts of an input text or image.
- ▶ Generative model training: entails learning to generate new data
- ▶ Pretrained models: A pretrained LLM and a pretrained vision encoder can be used, with an added mapping network layer that aligns or projects the visual representation of an image to the LLM's input space.

Ambiguity resolution

PDDL planning

PDDL problem generation experiments